Capacity, departure scheduling and congestion in MATSim

Developing and evaluating scenarios with the official equil network
LSGI3213: Intelligent Transportation Systems
Computational project report | 8 September 2026

Abstract

This project examines how road capacity and departure scheduling influence travel in a synthetic agent-based network. Public, version-tagged MATSim 12.0 equil inputs were downloaded and verified against the tutorial distribution. The original 100-agent example was reproduced, and the supplied 2,000-agent population was used to define a separate congestion benchmark. Five main scenarios compare unchanged supply, doubled capacity on nine corridor links, departures spread symmetrically across one hour, their combination, and a configuration allowing departure-time learning. Each was executed for iterations 0-100 with three random seeds.

All 4,000 planned car trips completed in every main run, with zero stuck events. Across seeds, mean morning travel time was 22.05 minutes in the benchmark, 18.60 after expansion, 15.00 with staggering, 15.00 with both, and 15.01 with adaptive timing. These changes correspond to reductions of 15.7%, 32.0%, 32.0% and 31.9%, respectively. Trip distributions, executed-plan scores and event-derived link conditions reveal both average benefits and differences among agents.

The exercise demonstrates a reproducible ITS scenario workflow, rather than a forecast for a real city. Its main limitations are synthetic demand, fixed activity locations, car-only travel, simplified queue dynamics and a small number of stochastic replications. Plan scores are model-specific utility measures; they are not monetary welfare. A complete replication package contains downloaded inputs, scenario XML, run logs, event files, derived results and figure/report source code.

Keywords: MATSim; equil; traffic simulation; departure scheduling; capacity; intelligent transportation systems

1 Background and research questions

A transport network can experience congestion even when the total number of travellers remains unchanged. The concentration of departures and the allocation of those travellers across routes determine how quickly demand reaches individual road links. Capacity improvements act on supply, while departure scheduling acts on when demand arrives. An ITS analysis should therefore inspect both traffic performance and the behaviour generating that traffic, rather than reporting only a single network-wide average.

MATSim provides a useful controlled environment for this question because individual agents execute activity plans and experience congestion created by other agents. The equil network offers several parallel outward corridors with a shared return path. Its small size makes the link-level mechanism inspectable. Choosing this example also preserves the assignment requirement to start from equil or chessboard; an unrelated metropolitan road download would not directly satisfy that requirement [1,2].

Questions addressed

RQ1: How much does doubling the discharge capacity of the nine outward corridor links reduce morning travel time and alter their use?

RQ2: Can a one-hour spread of planned departures reduce congestion without changing the number of travellers or their activity locations?

RQ3: Does combining both interventions provide an additional benefit, or do they address the same queues?

RQ4: What changes when departure times are learned through the MATSim configuration rather than assigned externally?

The hypotheses are directional, not predetermined results: expansion is expected to shorten queues, staggering is expected to reduce their formation, and their joint gain may be smaller than the sum of separate gains. Timing adaptation may improve daily scores, but its score benefit need not equal its travel-time benefit because activity timing also enters the scoring system. The completed simulations, rather than these expectations, determine the findings.

This is a within-model intervention comparison. Common inputs and paired seeds improve comparability, but do not establish external validity. Neither the network coordinates nor the population represent observed Hong Kong residents. No observed counts, household survey or traffic-speed dataset is available for calibration. The analysis consequently makes conditional statements about the constructed experiment.

2 MATSim framework and mechanisms

MATSim executes individual daily activity plans on a transport network. A plan describes activities, locations, timing and the trips joining those activities. Network supply is represented by directed links with lengths, free speeds, lanes and capacities. The mobility simulation produces events such as departures, arrivals and link entries or exits. These events support both behavioural scoring and the performance indicators used in this report [3,4].

The repeated simulation cycle

In each iteration, agents select plans, execute them together, receive scores and may revise their plans for a subsequent iteration. In the main fixed-time scenarios, BestScore has weight 0.9 and ReRoute has weight 0.1. Agents retain at most five plans. BestScore chooses among remembered alternatives; ReRoute provides route innovation. Repeated interaction can improve experienced scores, although a finite simulation and a stable average do not prove convergence to an exact equilibrium [2,5].

QSim represents traffic using a queue-based network loading mechanism. A vehicle must spend at least the free-speed traversal time on a full link, then compete for discharge subject to flow and downstream constraints. This is not a detailed car-following simulation. Increasing the capacity attribute changes discharge flow but, in this experiment, does not add a physical lane or increase storage. Those distinctions limit the engineering interpretation of the expansion scenario [3,4].

Scoring and time choices

The supplied configuration uses a performing utility of +6 utils per hour, car-travel utility of -6 utils per hour and a late-arrival coefficient of -18 utils per hour. Typical home and work durations are 12 and 8 hours. Work opens at 07:00, closes at 18:00 and has a latest start time of 09:00. The full activity utility is nonlinear and includes timing conditions: these coefficients do not imply that the final plan score is simply a linear function of travel minutes [2,5].

The adaptive scenario replaces 0.1 of the BestScore weight with TimeAllocationMutator. Its mutation range is 1,800 seconds and mutationAffectsDuration is false, so the explicit work duration is retained while eligible activity end times can change. Repeated mutations can accumulate across iterations. Therefore, the mutation range is a per-change limit, not a permanent one-hour departure window [6].

3 Public data and provenance

The assignment mentions input ZIP files hosted on Blackboard. Those specific ZIP files were not accessed. Instead, five inputs were retrieved from the official matsim-org/matsim-libs repository at the matsim-12.0 tag. All five downloaded files were byte-identical to the corresponding files already present inside the local Mac tutorial distribution. This is direct evidence of agreement with the tutorial examples, but it cannot prove identity with an unseen Blackboard project ZIP [1,2].

Public file	Role and audit result
network.xml	15 nodes; 23 directed links; synthetic coordinates
facilities.xml	23 facilities with home/work activity options
config.xml	Original 100-agent example and scoring settings
plans100.xml	100 people; 201 planned trips; source reference
plans2000.xml.gz	2,000 people; home-work-home; 4,000 trips

The repository inputs are synthetic demonstration data. They are suitable for the chosen controlled scenario but must not be labelled as measured traffic. The optional counts file was not used as a calibration dataset. Each downloaded file has a saved URL, retrieval timestamp, GitHub blob identifier, SHA-256 digest and byte count in data/source_manifest.json. The local comparison is recorded independently in data/local_comparison.json.

The 2,000-agent file contains legacy v4 route descriptions. For the five congestion scenarios, those embedded routes and departure-time hints on legs are removed consistently, allowing MATSim to generate routes on the retained network. Activity links, facilities, modes and work duration remain unchanged. This preprocessing is applied to the benchmark as well as every intervention; it is not a treatment-specific advantage. Raw downloads are retained separately and never rewritten.

The source 100-person file is retained without this preprocessing. It includes one extra same-location work-to-work trip, and its work schedules differ from the 2,000-agent file. Its result is therefore reported as a reproduction check, not used to estimate the effect of multiplying population size. The five main scenarios all use the same 2,000 people and differ only in the declared interventions.

4 Base network and daily activities

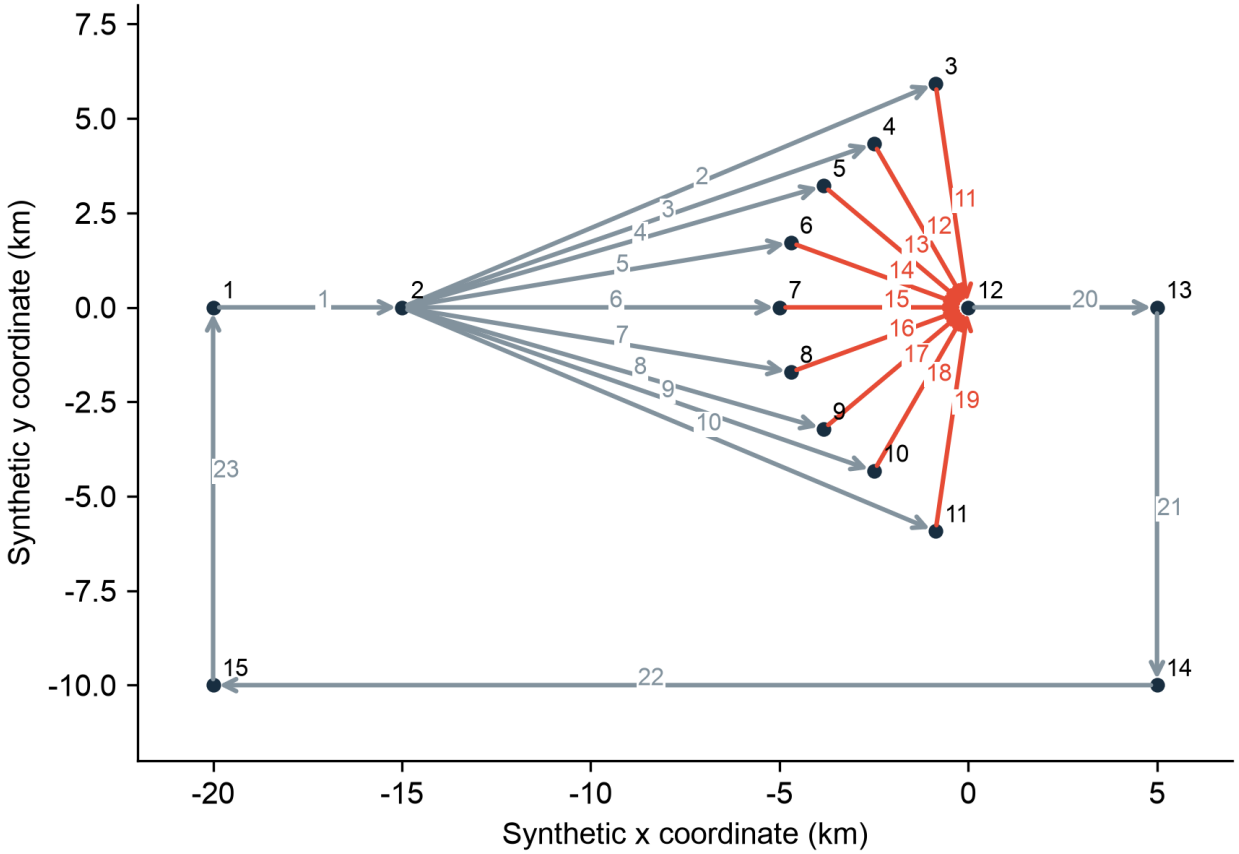


Figure 1. Official equil topology. Node and link labels are identifiers; red links 11-19 are the capacity treatment. The coordinates show a synthetic layout, not a geographic map. Link lengths are taken from XML and need not equal straight-line distances.

All 2,000 agents start at activity link 1, travel to work at link 20, perform an eight-hour work activity and return home. In the congestion benchmark, first departures are all scheduled at 06:00. Nine outward alternatives connect node 2 to node 12 through nodes 3-11. Their upstream links 2-10 have capacities of 3,600 vehicles per hour, while links 11-19 each provide 1,000 vehicles per hour. The latter are the deliberately targeted corridor bottlenecks.

The nine treatment links each have a declared length of 5 km, one lane and free speed of 27.78 metres per second. Their combined nominal flow capacity is 9,000 vehicles per hour. Other connector and return links have much higher flow capacities. The return path is longer than the outward path, so morning and return journeys must be separated when interpreting daily averages. Endpoint treatment also means that dividing the sum of every XML link length by speed does not automatically reproduce trip duration.

Facilities are supplied and referenced by the 2,000-person plans. Their locations are held fixed to isolate route and timing responses. Neither residential relocation nor destination choice is enabled; differences in outcomes cannot be interpreted as evidence about longer-term land-use adjustment.

5 Scenario design and justification

Scenario	Network / daily-plan change	Configuration
Baseline	2,000 source agents; all depart 06:00	BestScore 0.9; ReRoute 0.1
Capacity	Links 11-19: 1,000 to 2,000 veh/h	Same as baseline
Stagger	First departure spread 05:30-06:30	Same as baseline
Combined	Both capacity and staggering	Same as baseline
Adaptive	Same initial inputs as baseline	BestScore 0.8; ReRoute 0.1; time mutation 0.1

The capacity scenario changes only discharge capacity on links 11-19. Link lengths, free speeds, lanes and all other link capacities are fixed. It tests operational flow capacity, not a fully costed road-widening proposal. The stagger scenario assigns deterministic midpoint quantiles across a 3,600-second interval: for zero-based person position i, departure equals 19,800 + (i + 0.5) x 3,600 / 2,000 seconds. The mean stays exactly 06:00 in the XML; one-second QSim discretisation may slightly alter realised times.

The staggered schedule is assigned by source-file order and kept identical across seeds. It is a controlled, perfect-compliance scheduling experiment. It does not estimate whether workers would accept earlier departures. The work activity still lasts eight hours from arrival, so return departure times can change even though work duration does not. The combination scenario permits a factorial comparison of overlapping benefits.

The adaptive scenario supplies a distinct configuration intervention. It retains baseline network supply and initial schedules, but permits endogenous timing changes under the unchanged scoring rules. Its outcome is compared with externally staggered schedules to illustrate the difference between assigned coordination and individual score-seeking behaviour. Because early work arrivals interact with the 07:00 opening time, a timing policy can affect scores beyond its effect on queues.

All main scenarios share the same scoring function, facility data, car mode, demand size, memory limit and iteration horizon. Increasing the horizon from the original 10 to 100 iterations is a common experimental control, rather than an intervention whose travel benefit is attributed to a policy.

6 Execution and measurement

The project uses the MATSim 12.0 executable supplied in the tutorial and Amazon Corretto Java 11.0.30 on this Mac. The JAR manifest identifies a Java 11 build. The tutorial suggests Java 14, but actual successful execution with Java 11 is documented in the saved logs. All simulations are run headlessly with a 2 GB maximum heap and one thread for QSim, global processing and event handling. No global Java settings or shell profile were changed.

Control	Value / interpretation
Main replications	5 scenarios x 3 seeds = 15 runs
Seeds	4711, 42, 2025; paired across scenarios
Iteration horizon	0-100 inclusive: 101 simulated days per run
Reference run	Original 100-person input; iterations 0-10
Evidence	Exit codes, console logs, output configuration, plans, events and native trip tables

Final-iteration event files are the primary measurement source. A trip begins with a car departure event and ends with its matching arrival. Trip duration is arrival minus departure. Stuck trips, unmatched departures and unmatched arrivals are checked explicitly. The independent event-derived durations are matched by person and trip number to every row of MATSim output_trips.csv.gz. Selected-plan mean scores are independently checked against the final avg. EXECUTED value in scorestats.txt.

Full-link traversal time is measured between entered-link and left-link events for the same vehicle and link. Delay is the nonnegative excess over XML length divided by free speed. Exit flow is exits per bin, scaled to an hourly rate. Occupancy integrates the vehicle count over time; density divides mean occupancy by link kilometres and lanes. Origin and destination activity links involve partial traversals. Traversal and flow measures for links 1 and 20 are marked missing; event-based occupancy and density are retained.

Results use both 5-minute and 15-minute bins for traffic concentration. A flow/capacity ratio is measured throughput relative to nominal flow capacity, not unmet demand. Empty traversal bins are missing observations rather than zero travel time. Cross-seed ranges describe simulation sensitivity, not confidence intervals for a real population.

7 Trip totals and travel performance

Scenario	Trips / run	AM mean (min)	AM p95 (min)	Daily travel (h)
Baseline	4,000	22.05	29.31	2035.1
Capacity	4,000	18.60	22.36	1920.0
Stagger	4,000	15.00	15.02	1800.1
Combined	4,000	15.00	15.00	1800.0
Adaptive	4,000	15.01	15.05	1800.4

Table 4. Scenario results are arithmetic means of three final-iteration run summaries. AM p95 is the mean of the three within-run 95th percentiles, not a pooled percentile. Daily travel hours sum all completed trips in a run. Each run contains 2,000 first trips and 2,000 return trips, with zero stuck agents or unmatched trip events.

The capacity change lowers mean morning travel time by 3.45 minutes per agent (15.7%). Staggering lowers it by 7.05 minutes (32.0%). The combination lowers it by 7.05 minutes (32.0%). These comparisons hold demand size and activity locations fixed. They measure simulated time savings, not reductions in the number of journeys.

The adaptive configuration yields a morning average of 15.01 minutes. Its daily mean across both directions is 27.01 minutes, compared with 30.53 in the baseline. Reporting both quantities avoids obscuring the morning response with the longer, high-capacity return journey. The event analysis retains all completed trips rather than excluding slow trips to improve the result.

The separate original-data run completes 201 trips for 100 agents and yields a morning mean of 15.55 minutes and a mean executed score of 65.09 utils. These values confirm a working example, but they are not the denominator of the treatment effects above. The source populations have different plan structures and schedules; their absolute scores should not be treated as a controlled population-size comparison.

8 Departure patterns and distributions

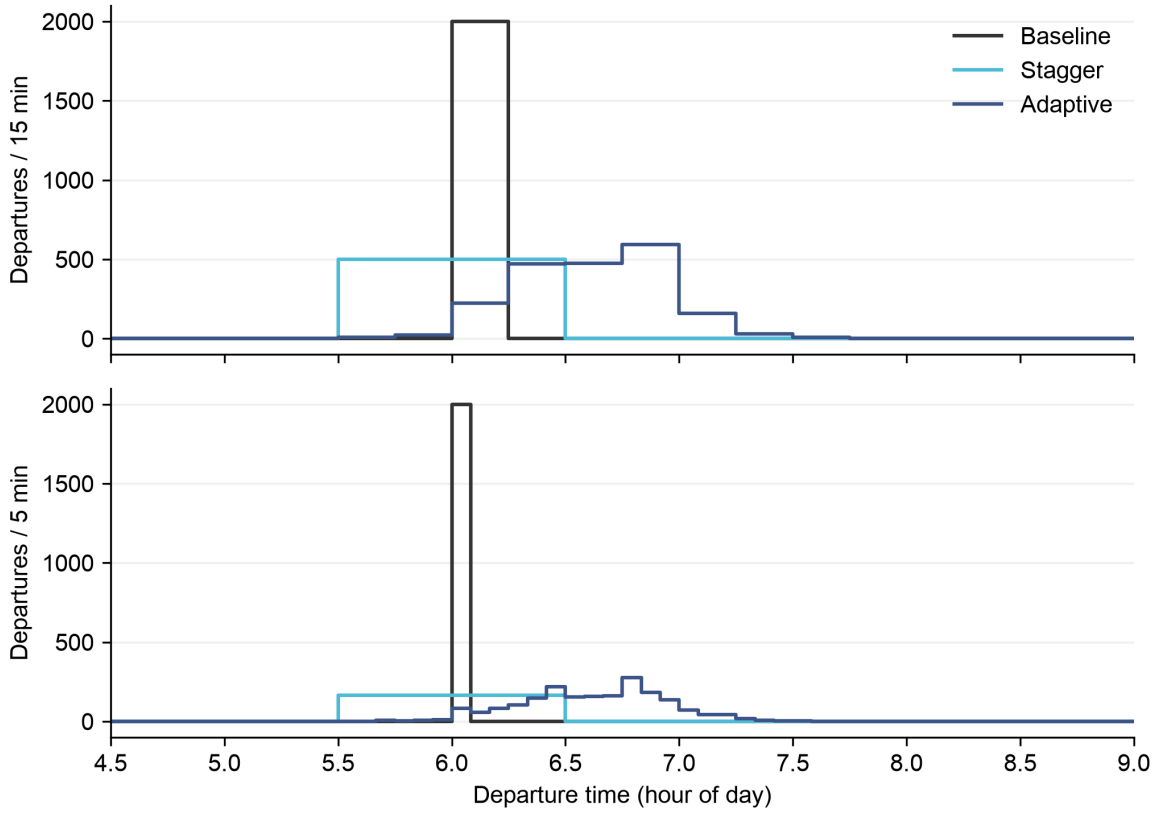


Figure 2. Realised first departures for seed 4711, shown with 15-minute and 5-minute bins. The baseline begins as a sharp pulse; external staggering and time learning redistribute departures. Capacity shares baseline initial timing, and Combined shares Stagger timing.

Baseline departure times are concentrated at 06:00. Staggering retains that mean but spreads the input schedule uniformly. Adaptive timing has a mean realised first departure of 6.636 hours and a within-run departure-time standard deviation averaging 19.46 minutes. This spread is an outcome of repeated replanning, not a directly imposed uniform schedule.

The 5-minute view preserves short pulses that a 15-minute histogram can hide. Counts are displayed per bin, so their numerical heights are not directly comparable across the two resolutions. The event-derived departure time is an activity-end decision; a simultaneous departure does not mean that all vehicles immediately enter the first full road link. Insertion and downstream discharge still constrain physical movement.

Morning travel-time distributions are retained in the accompanying ECDF figure and trip data. A lower mean is not sufficient to conclude that every agent benefits. The final comparison therefore also pairs individual people across scenarios and reports the spread of their time changes.

9 Daily plan scores and learning

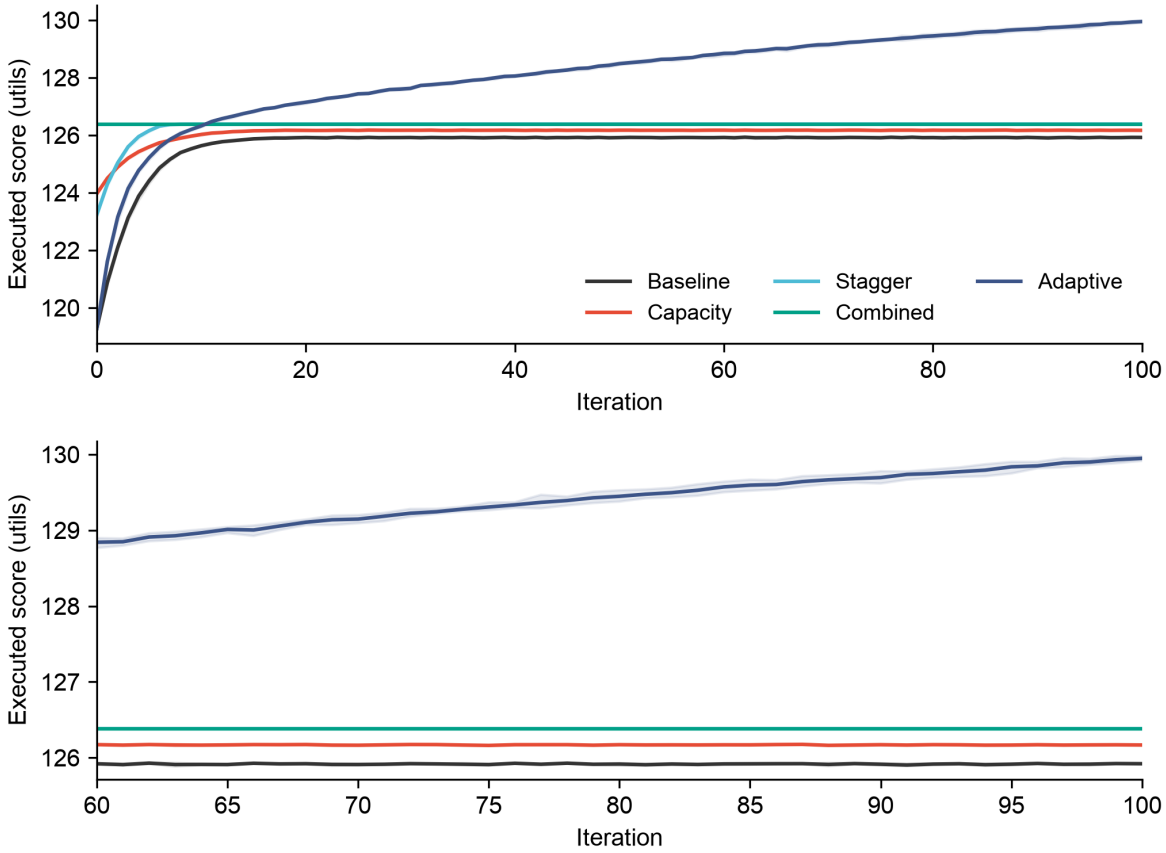


Figure 3. Mean executed score by iteration. Lines average three runs; shading is the minimum-maximum range across seeds. The lower panel magnifies iterations 60-100. Scores use identical scoring settings across the five main scenarios.

The final mean executed score is 125.92 utils for Baseline, 126.17 for Capacity, 126.38 for Stagger, 126.38 for Combined and 129.95 for Adaptive. Executed score is used because it describes plans experienced that iteration. The best score held in plan memory can reflect an earlier traffic environment and is not a substitute for current performance.

The adaptive alternative changes both travel experience and activity timing. A larger score gain cannot be converted directly into monetary welfare or attributed wholly to congestion reduction. With 07:00 work opening, some early arrivals cannot obtain the same activity utility as later arrivals. Fixed-duration work and the unchanged scoring function keep the comparison interpretable, but they do not remove timing preferences from the result.

10 Route allocation and link states

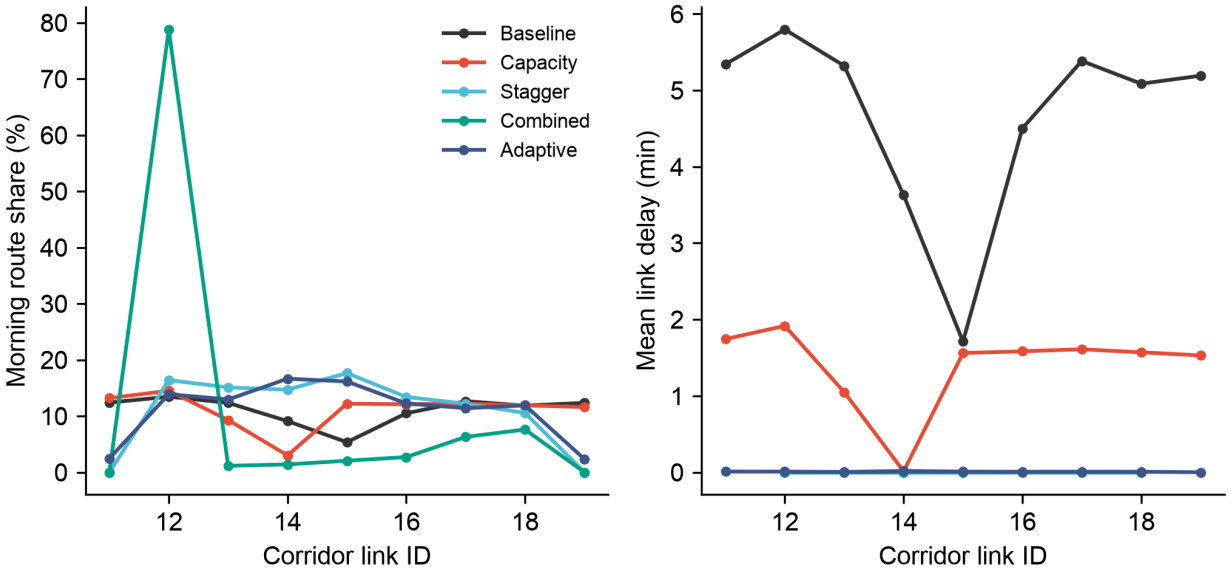


Figure 4. Final route shares and mean full-link delays on links 11-19, seed 4711. Each outward trip traverses exactly one of these corridor links. Route shares total 100%; link delay excludes the free-speed traversal component.

Route shares reveal how agents distribute themselves across the nine corridors. Since all outward alternatives have the same declared lengths and free speeds, queues and score history can drive different shares without a geometric distance advantage. Low-congestion scenarios can preserve an uneven allocation when alternatives offer almost identical scores. Equal route use is therefore not a necessary condition for good network performance.

Mean corridor-link delay is 5.02 minutes in Baseline, 1.52 in Capacity, 0.00 in Stagger and 0.00 in Combined, averaged across seeds. Delay is measured on the actual full-link traversal and compared with 5,000 / 27.78 seconds of nominal free-speed time. It includes queue waiting and QSim discretisation, so small positive residuals need not indicate a substantial queue.

Doubling capacity leaves one lane per treatment link. Thus, the intervention changes the rate at which queues can discharge, while the density denominator remains unchanged. A faster queue clearance can raise a short-bin exit flow even while mean travel time improves. That combination is not contradictory: throughput measures movement, while travel time measures the duration experienced by a vehicle.

11 Traffic dynamics through the peak

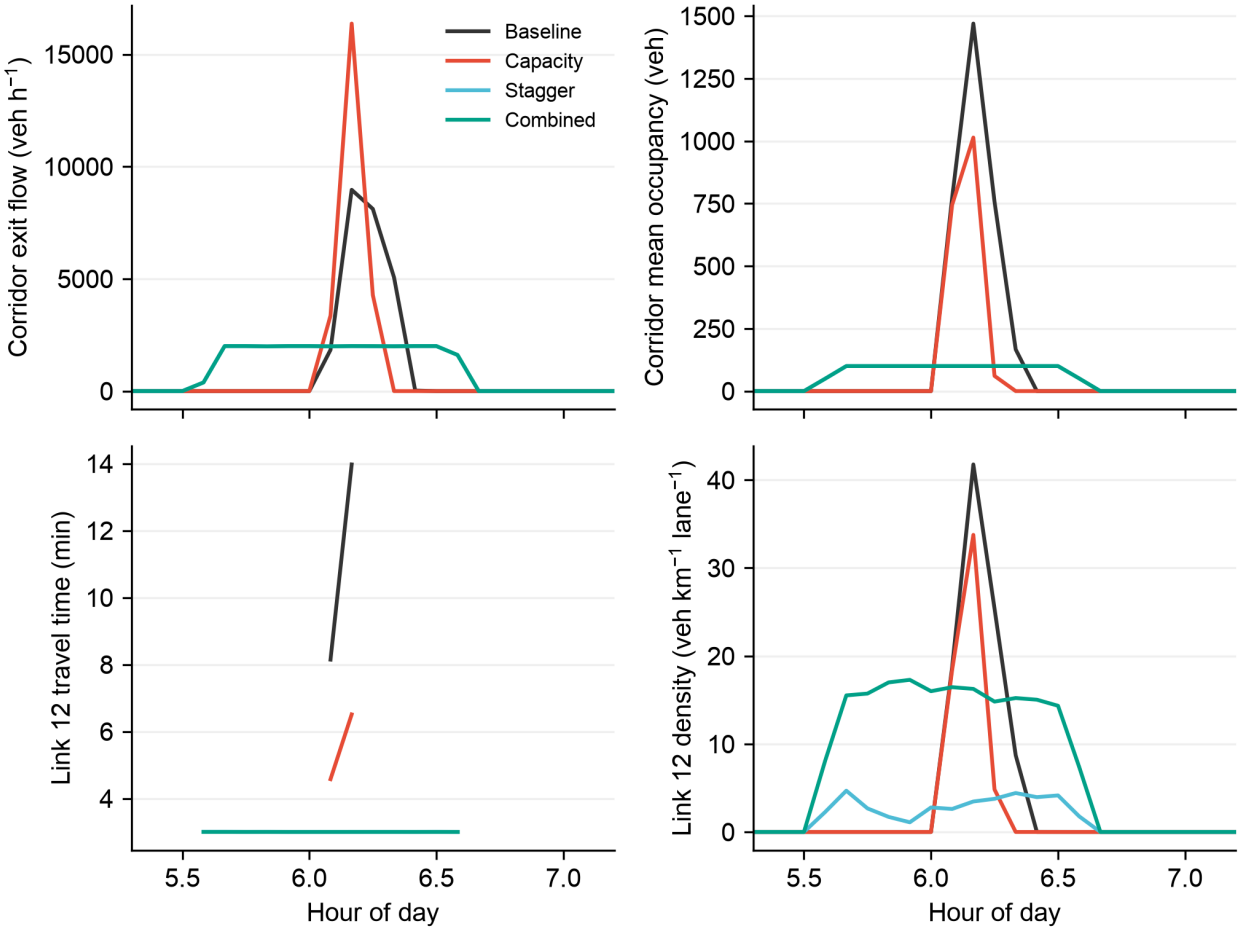


Figure 5. Five-minute link-state summaries for seed 4711. Top: summed exit flow and time-averaged occupancy on the nine corridor links. Bottom: full traversal time and density for link 12. Time bins are shared across scenarios.

Occupancy and exit flow answer different questions. Summed occupancy records how many vehicles are on the corridor links on average during a time interval. It includes both moving and queued vehicles, so it is not a direct queue-length observation. Exit flow records completed link traversals, expressed as an hourly rate. A high exit flow may describe successful clearance of a queue rather than a worsening condition.

The lower panels show a specific link rather than only the aggregate. Mean travel time groups completed traversals by entry time. Exit-flow bins group by exit time, so their peaks need not coincide. Density is calculated by exact integration of entry and exit events and division by 5 km and one lane. These definitions prevent a misleading direct comparison between a snapshot vehicle count, a bin-average density and a trip-weighted speed.

The pattern is consistent with the intended mechanisms: capacity changes the service rate; staggering changes the arrival pattern. Because both influence the same morning queues, their benefits can overlap. Link-level evidence should accompany the scenario mean when explaining why a combined intervention adds relatively little or substantially more than an individual measure.

12 Spatial and individual heterogeneity

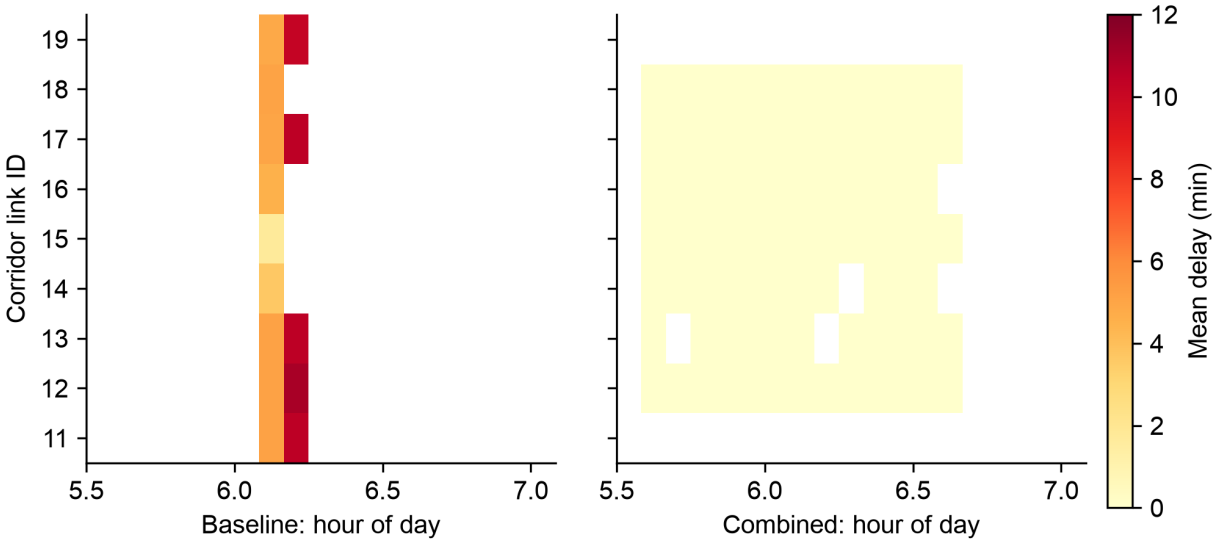


Figure 6. Corridor delay by link and entry-time bin for Baseline and Combined, seed 4711. Both panels use the same 0-12 minute colour scale. White cells have no completed traversal entering that bin; they are not evidence of zero delay.

The heatmap preserves the identity of each treated link and the timing of its experienced delay. A network average can hide a badly performing route or conceal that a queue has merely shifted in time. Empty observations remain visibly separate from low-delay observations. The companion numerical link tables allow inspection without estimating values from colour.

The individual comparison pairs each person identifier between the baseline and an intervention at the same seed. Its outcome is baseline minus intervention morning duration; a positive value is a time saving. This pairing preserves the identity of an agent, but not necessarily their selected route or queue position. A small group can consequently experience a loss even when the total travel-time change is favourable.

Despite higher mean scores, 41.2-41.6% of agents have a lower executed-plan score under Stagger and Combined than under the paired baseline, across the three seeds. This measured tradeoff reflects scheduling as well as travel and cautions against equating average time savings with universal benefit.

No demographic variables are available. The analysis therefore describes heterogeneity in simulated experiences, not income, age or neighbourhood equity. The file-order assignment used for staggering is also not a realistic allocation rule for flexible work hours. A field study would need to measure schedule constraints and participation before treating this schedule as an implementable policy.

13 Robustness and remaining uncertainty

Scenario	AM mean range (min)	Score tail change (utils)
Baseline	21.935 - 22.145	-0.0003
Capacity	18.555 - 18.632	-0.0011
Stagger	15.002 - 15.002	-0.0000
Combined	15.000 - 15.000	+0.0000
Adaptive	15.010 - 15.012	+0.2451

Table 5. The range spans three random seeds. Tail change is the mean executed score over iterations 91-100 minus its mean over iterations 81-90, averaged across seeds. It is a diagnostic of recent drift, not a statistical test of convergence. Replanning remains active, and small changes in selected plans can persist after the average has flattened.

An extended baseline run at seed 4711, through iteration 200, gives morning travel time 21.931 minutes versus 21.935 at iteration 100. Its executed score changes from 125.929 to 125.929 utils; the extended tail score change remains +0.0041 utils. This is an independent rerun from the same initial inputs with a longer horizon, not an average over late-iteration event files.

An extended adaptive run at seed 4711, through iteration 200, gives morning travel time 15.007 minutes versus 15.010 at iteration 100. Its executed score changes from 129.915 to 131.004 utils; the extended tail score change remains +0.0368 utils. This is an independent rerun from the same initial inputs with a longer horizon, not an average over late-iteration event files.

The seeds are paired across scenarios, but their random streams can diverge when a scenario changes available actions. Pairing is useful for controlled bookkeeping and is not a guarantee of identical random draws at every behavioural decision. Three replications give only a limited view of algorithmic variability; the report shows ranges rather than presenting narrow confidence intervals as population evidence.

Five-minute and 15-minute link tables are both retained. Binning changes peak reported flow and density but cannot change the total completed-trip count or total travel time. Further sensitivity work should vary demand size, capacity increments, scheduling compliance, plan-memory settings and scoring parameters. These are proposed extensions, not experiments claimed to have been performed.

14 Interpretation for ITS planning

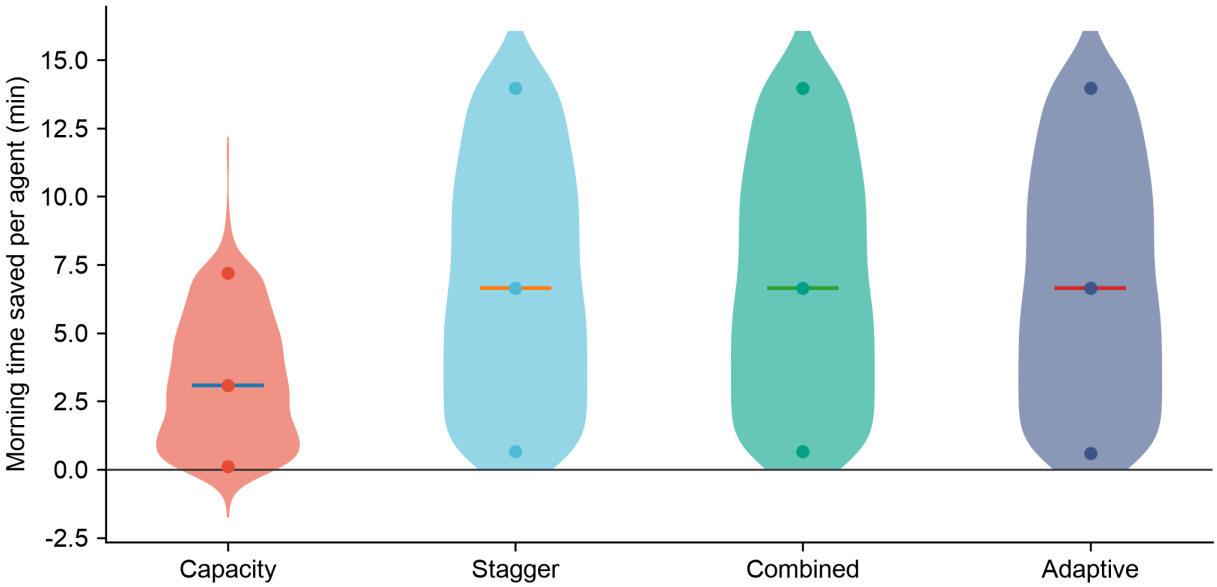


Figure 7. Paired morning time savings for 2,000 agents, seed 4711. Violins show the distribution; points mark the 5th, 50th and 95th percentiles. Positive values indicate a shorter trip than the baseline.

The factorial interaction in morning travel time is +3.45 minutes, computed as Combined - Capacity - Stagger + Baseline. A positive value means that the joint saving is smaller than the sum of the individual savings: the interventions partly remove the same delay. This arithmetic comparison describes the model response; it is not a regression-based causal estimate for an observed city.

An operational interpretation would compare measured bottleneck improvements with a scheduling programme. Capacity can be targeted to identified links, but construction cost, safety, emissions and implementation feasibility are not represented. Staggering avoids changing the network input, yet assumes full compliance and incurs unmodelled schedule inconvenience. Neither alternative can be labelled cost-effective from travel time alone.

The adaptive scenario illustrates how an ITS application that changes travellers' timing choices can influence network use. Actual advice, incentives or employer coordination are not explicitly simulated. Before recommending such a programme, the next step would be to calibrate observed schedules and route choices, represent partial participation, and assess distributional impacts alongside time savings.

15 Conclusions and reproducibility

The project successfully constructs, executes and compares MATSim scenarios derived from the official equil example. In the five main scenarios, every replication completes all 4,000 trips without stuck agents. Compared with the congested reference, capacity expansion reduces mean morning duration by 15.7%, staggering by 32.0% and their combination by 32.0%. The adaptive-timing configuration changes morning duration by 31.9% in the beneficial direction under the selected scoring assumptions.

Three findings are central. First, traffic concentration is a distinct modelling dimension from the number of daily trips: identical trip totals can generate different travel experiences. Second, plan scores and travel times must be analysed together, because timing and activity utility affect the former. Third, link flow, occupancy and traversal delay provide the mechanism behind a network-level result; none of those indicators alone is sufficient to label a link congested.

The results remain conditional on a synthetic network and demand set. The study has no observed validation data, no mode choice, no demand growth, no land-use response and no cost model. The capacity treatment does not add lanes, and the scheduling treatment assumes an externally assigned departure pattern. Additional random seeds and longer runs can test numerical robustness, but cannot resolve those structural limitations.

Reproduction and evidence

The project folder contains raw downloaded data, provenance hashes, generated scenario XML, MATSim outputs and all analysis code. scripts/build_scenarios.py constructs the inputs; scripts/run_scenarios.py executes them; scripts/analyze.py checks event and native statistics; scripts/make_figures.py regenerates the figures; scripts/make_report.py builds this report. The README gives environment and rerun instructions. The presentation deck and timed speaking notes accompany the report.

AI acknowledgment

OpenAI Codex assisted with public-data discovery, scenario design, code generation and execution, event analysis, figures, report text and presentation preparation during this project. This involvement extends beyond proofreading. The numerical results are produced by the saved MATSim runs, and the package includes the underlying evidence for independent checking. No personal contribution record, Blackboard submission or instructor approval is claimed [7].

References

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[5] MATSim developers (2020). MATSim 12.0 Java source and distribution. Relevant modules: org.matsim.core.mobsim.qsim, org.matsim.core.scoring, org.matsim.core.replanning and org.matsim.analysis. Local tutorial archive: matsim-12.0-sources.jar. Public source: https://github.com/matsim-org/matsim-libs/tree/matsim-12.0/matsim/src/main/java/org/matsim.

[6] MATSim developers (2020). TimeAllocationMutatorConfigGroup.java. mutationRange and mutationAffectsDuration parameter definitions. https://github.com/matsim-org/matsim-libs/blob/matsim-12.0/matsim/src/main/java/org/matsim/core/config/groups/TimeAllocationMutatorConfigGroup.java.

[7] OpenAI (2026). Codex assistance in the present computational project, 8 September 2026. AI contribution is disclosed in Section 15; executed simulation outputs, rather than generated prose, are the numerical evidence.

Data and figure attribution

Network and activity inputs originate from the official MATSim repository. All quantitative result figures were generated for this project from the saved simulation outputs. Figure source data, scripts and vector exports are included in the replication package. The supplied course and tutorial files remain unmodified.